

PrivML: Test Outputs

Timestamp: 2025-09-19 19:42:11

Model ID: Model 61

Dataset Path: data/Humans

Number of Images in Dataset: 7000

Training Size: 60%

Validation Size: 15%

Testing Size: 10%

Encrypted Testing Subset Size: 15%

Random Seed: 999

Plain Model Training

Epochs: 150

Batch Size: 16

Learning Rate: 0.005

Optimizer: Adam

Loss Function: MSELoss

Sample Output Count: 6

Plain Model Architecture

Hidden Layer Size: 512

Output Classes: 4

Conv Kernel Size: 1

Conv Stride: 2

Conv Padding: 0

Dropout Rate: 0.3

FHE Configuration

Encryption Scheme: CKKS

Scale (bits): 26

Polynomial Modulus Degree: 8192

Key Generation Level: 31

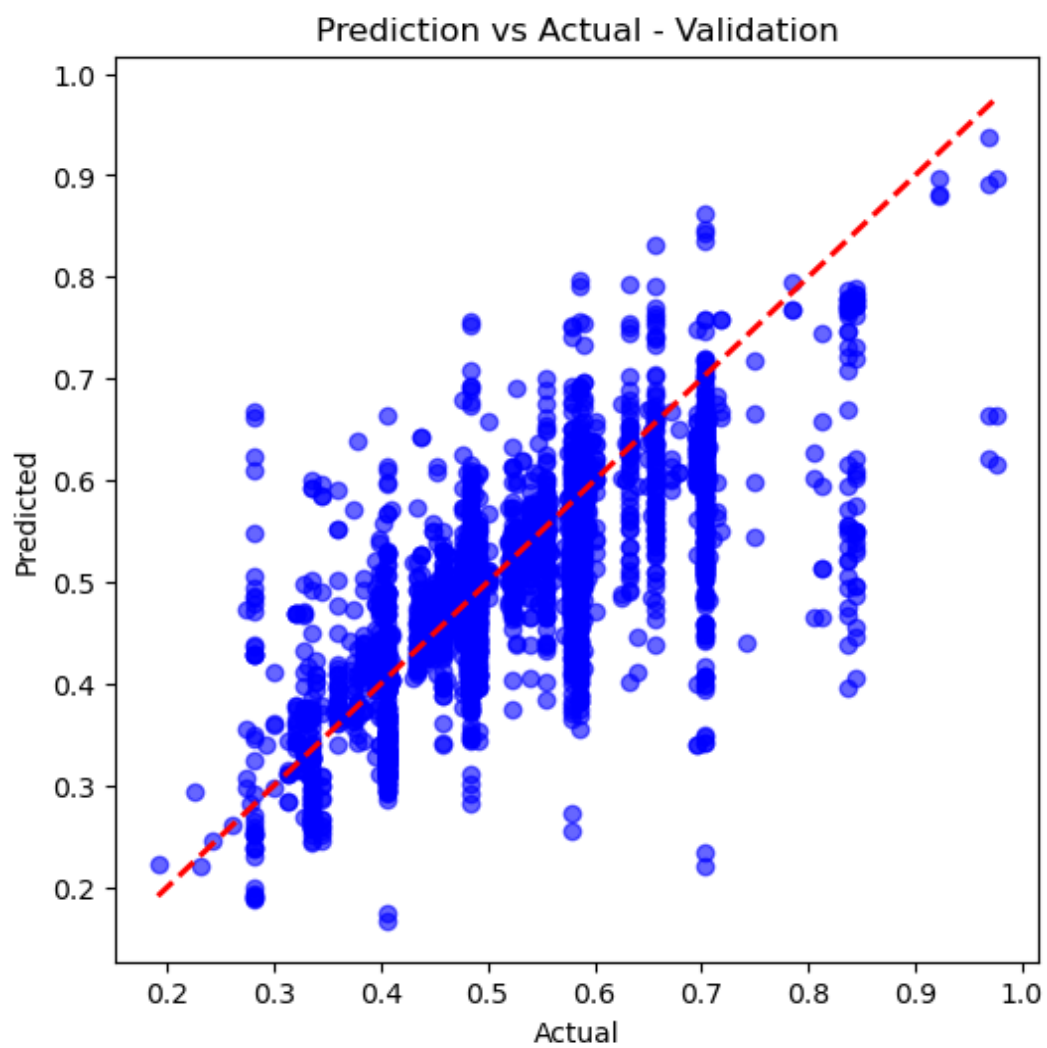
Decryption Stability Level: 31

Encrypted Loss Function: MSELoss

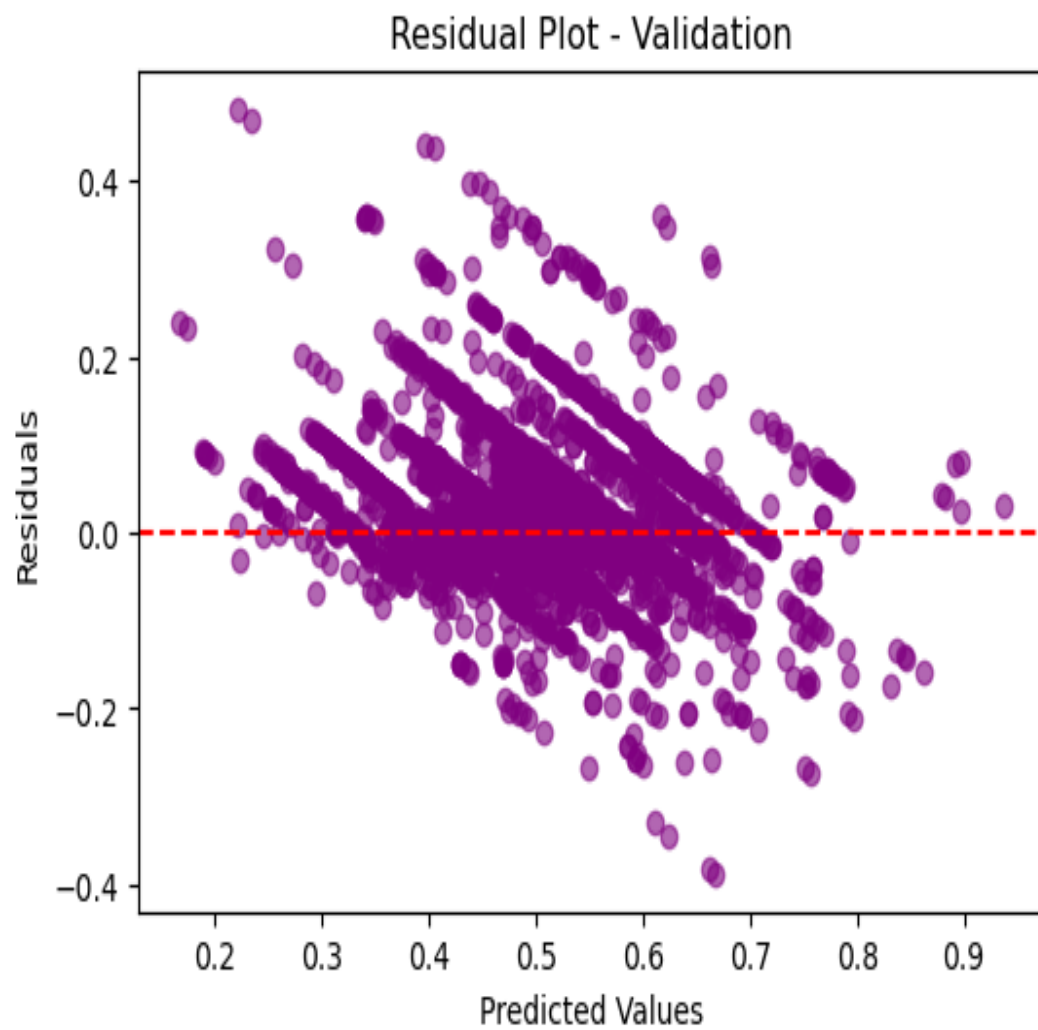
Coefficient Mod Bit Sizes: [31, 26, 26, 26, 26, 26, 26, 31]

Privacy-Preserving ML: ML Metrics Report

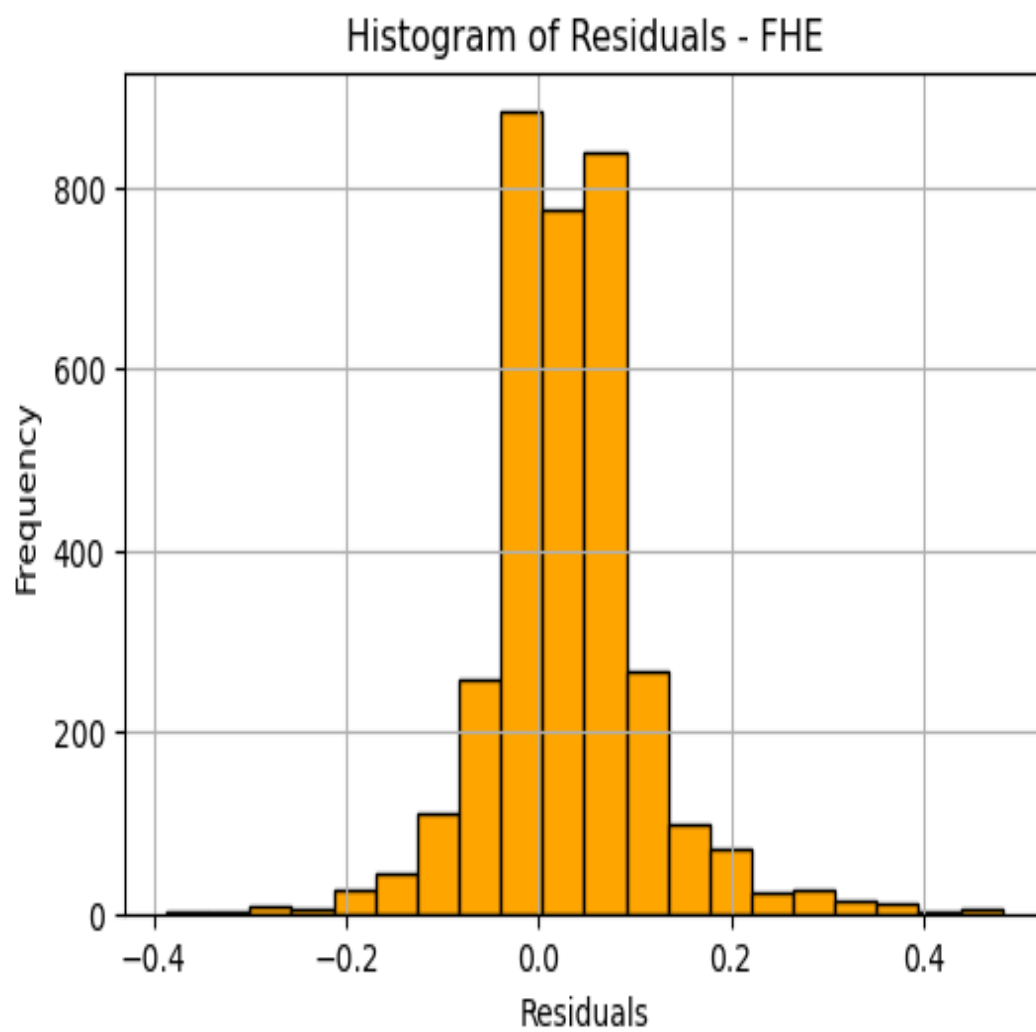
Prediction vs Plot - Validation



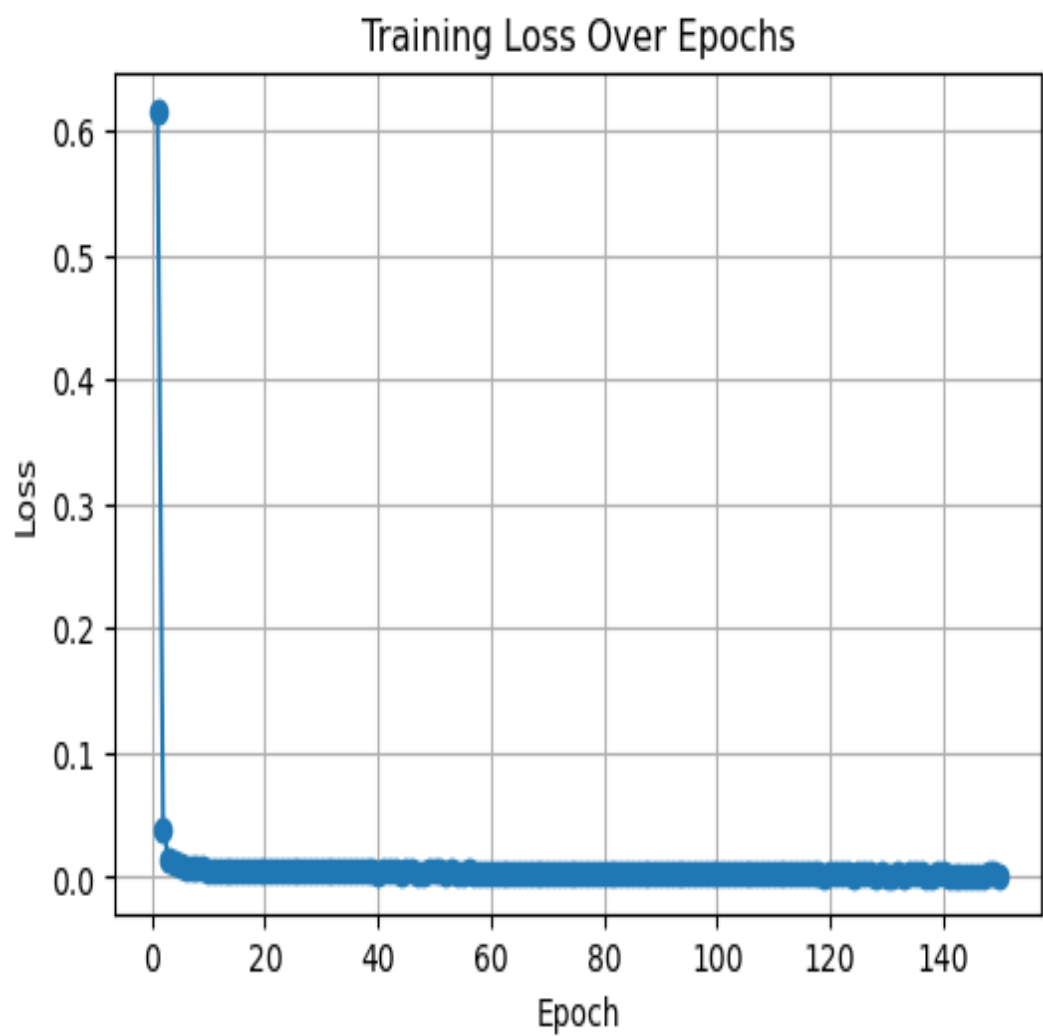
Residual Plot - Validation



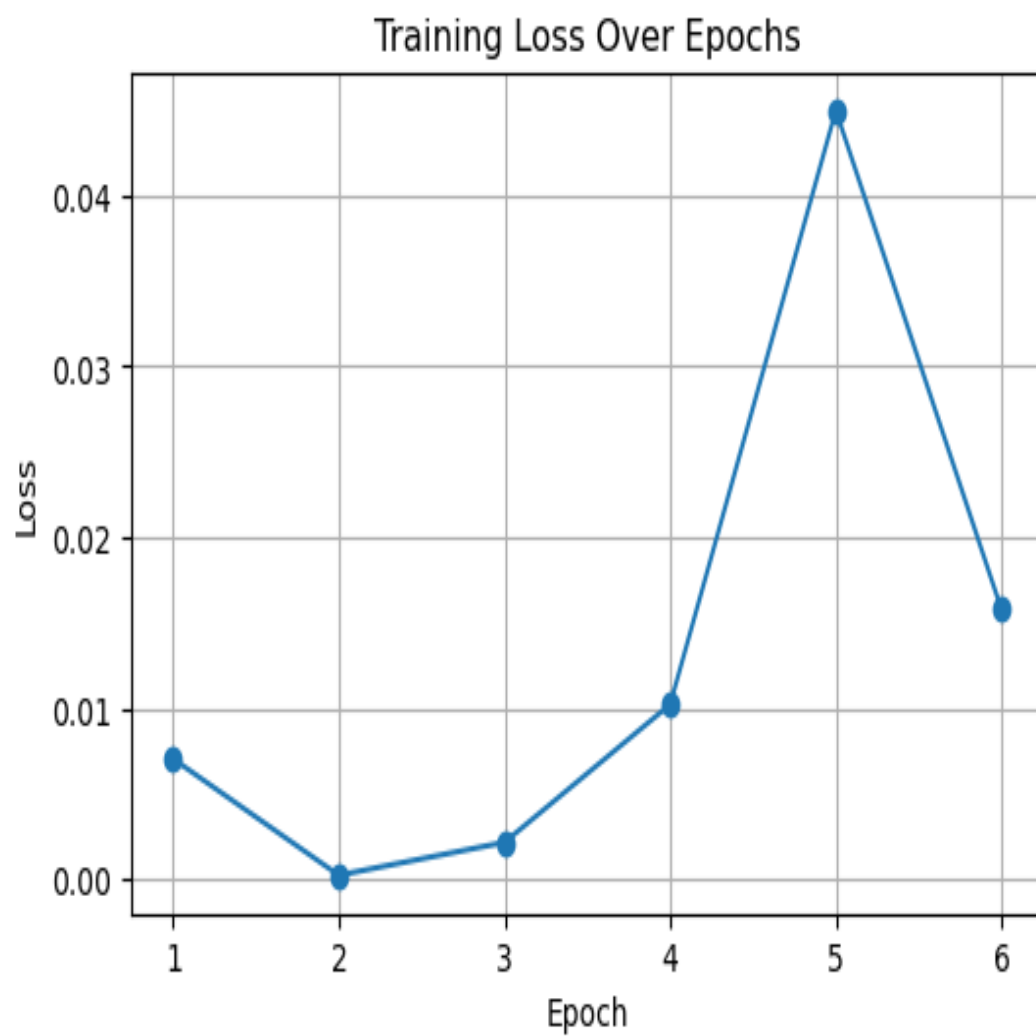
Residual Histogram - Validation



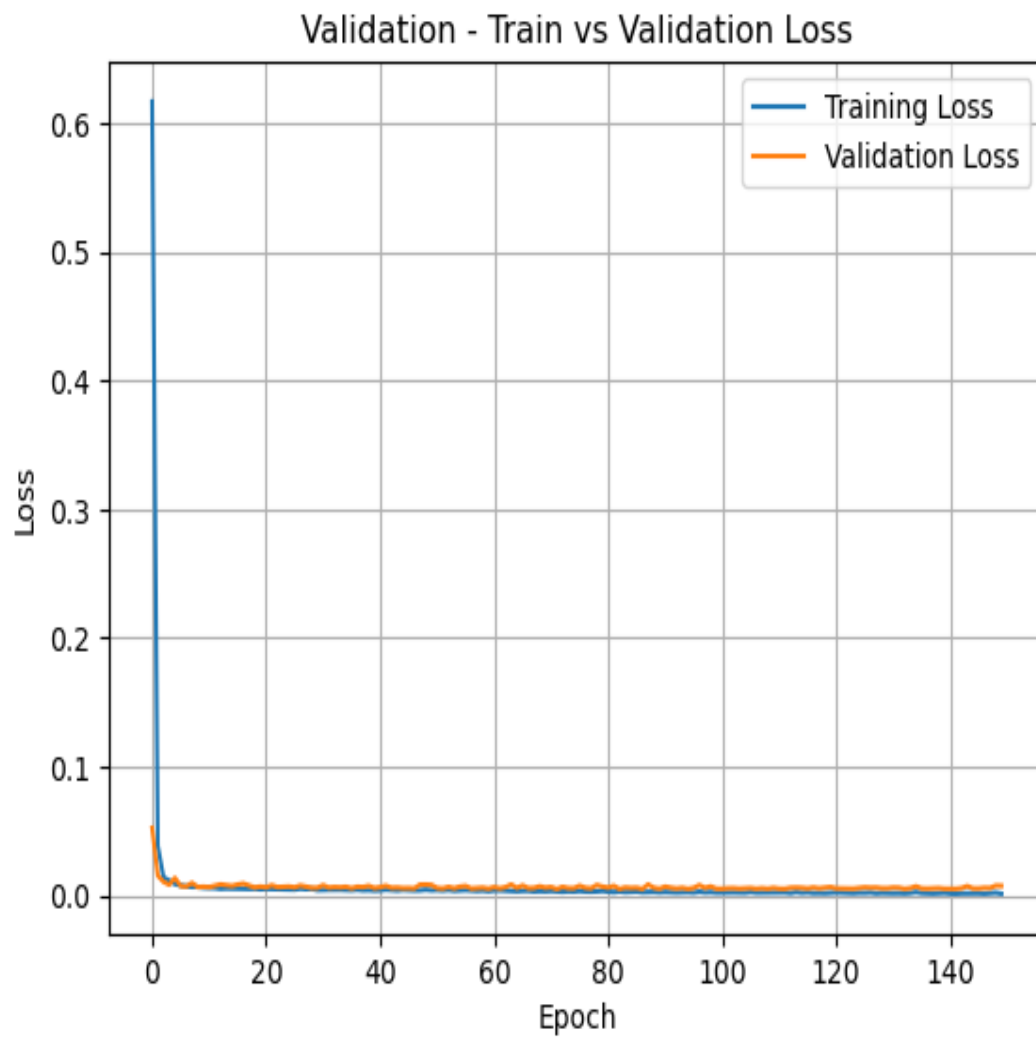
Training Loss - Validation



Training Loss - encrypted-fhe

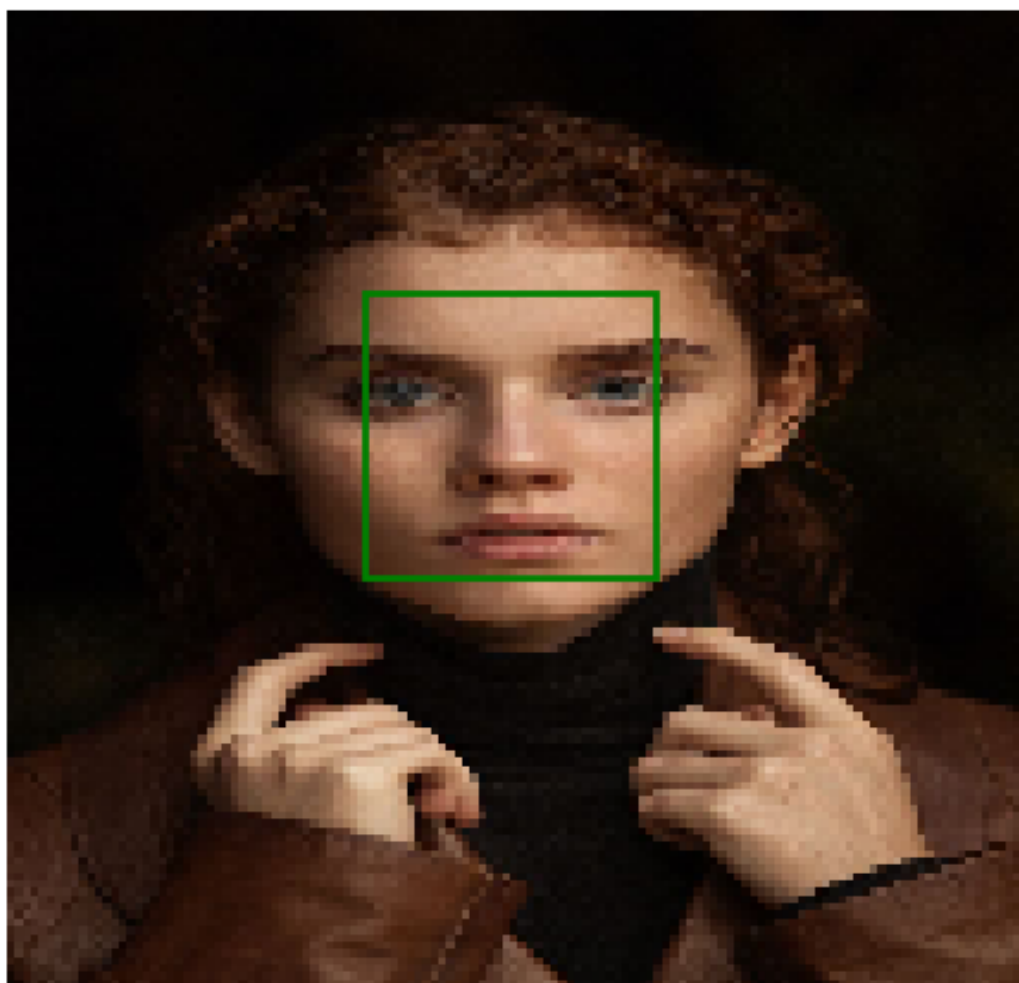


Train vs Validation Loss - Validation



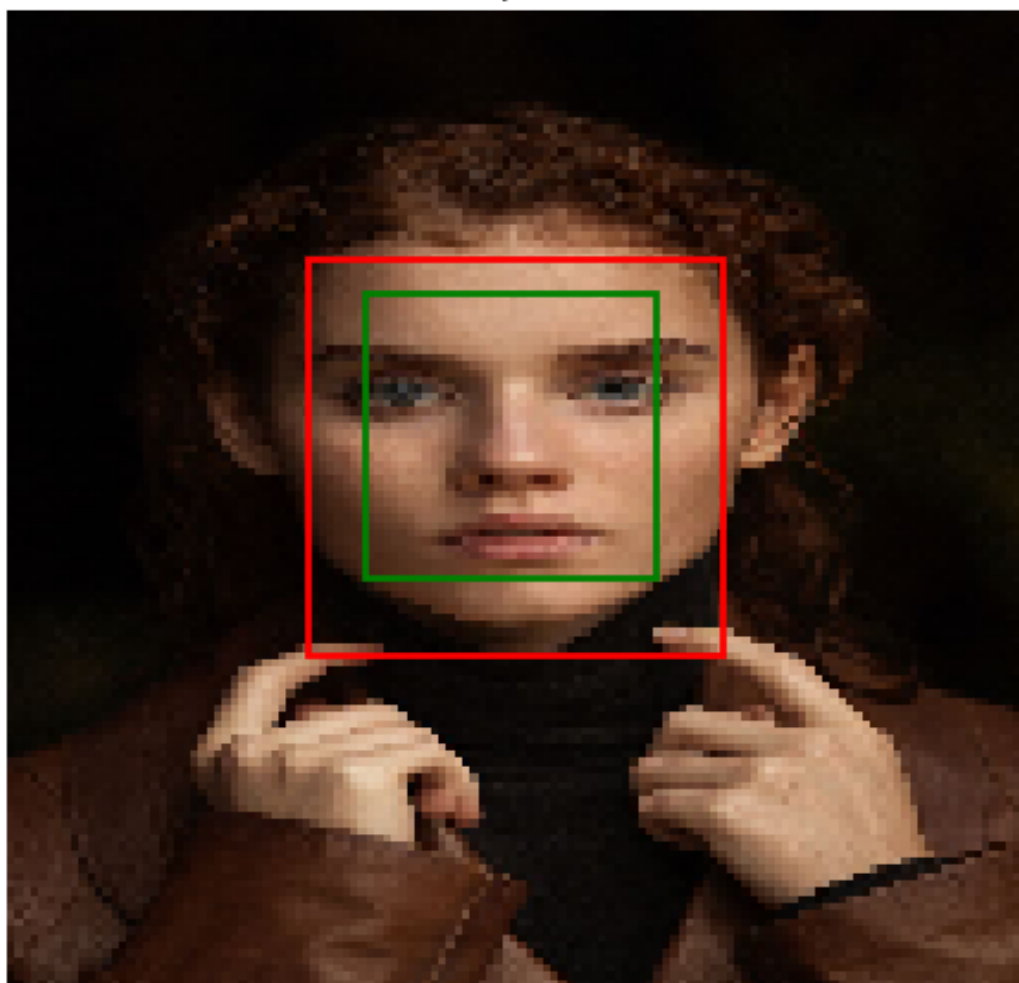
Plaintext: Model Prediction - Predicted Image

Green: Predicted



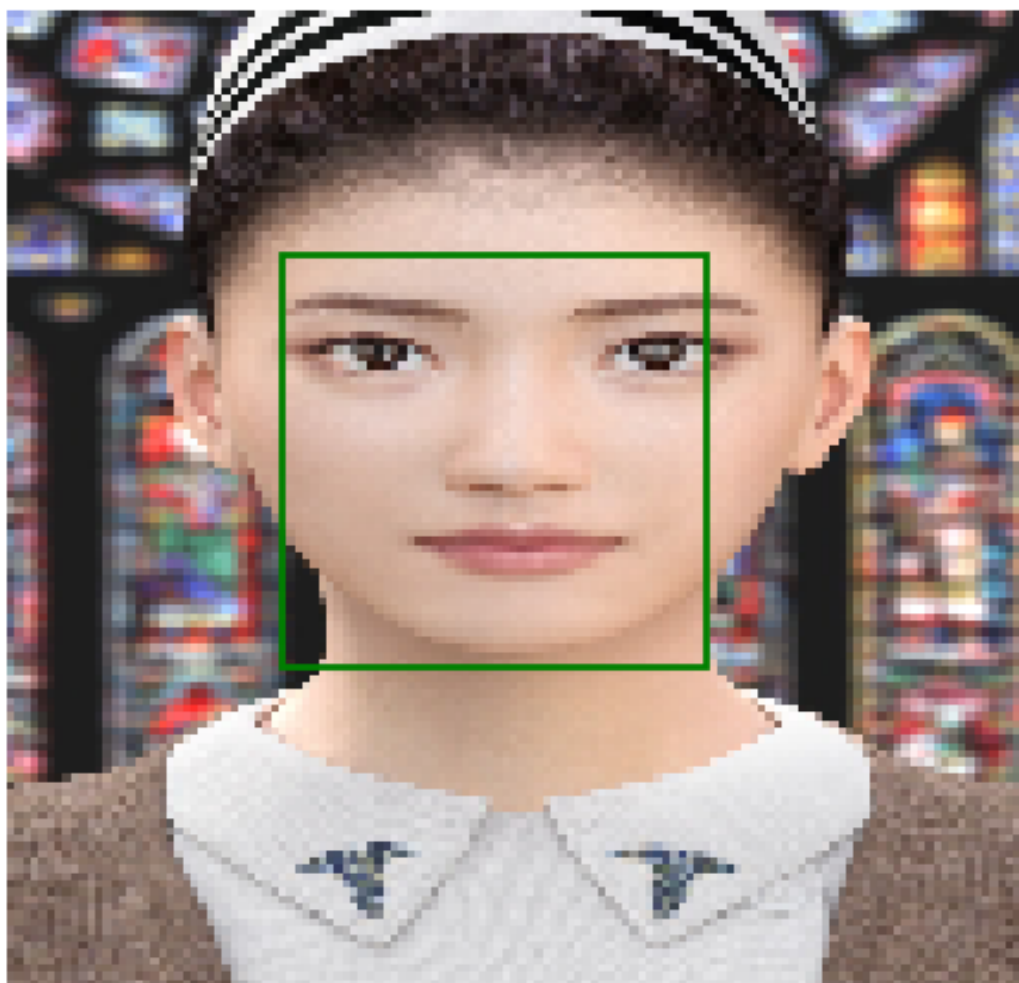
Plaintext: Model Prediction - Predicted Image

Green: Predicted | Red: Ground Truth



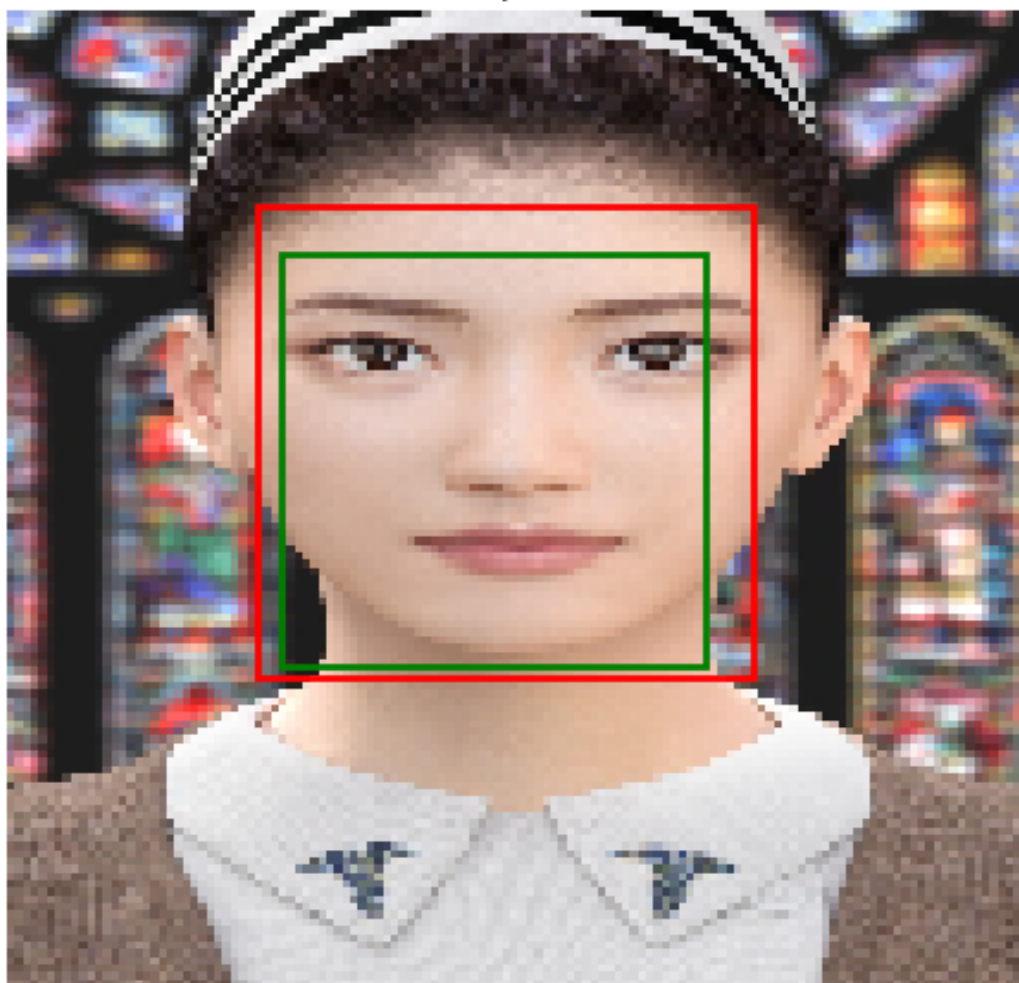
Plaintext: Model Prediction - Predicted Image

Green: Predicted



Plaintext: Model Prediction - Predicted Image

Green: Predicted | Red: Ground Truth



Plaintext: Model Prediction - Predicted Image

Green: Predicted



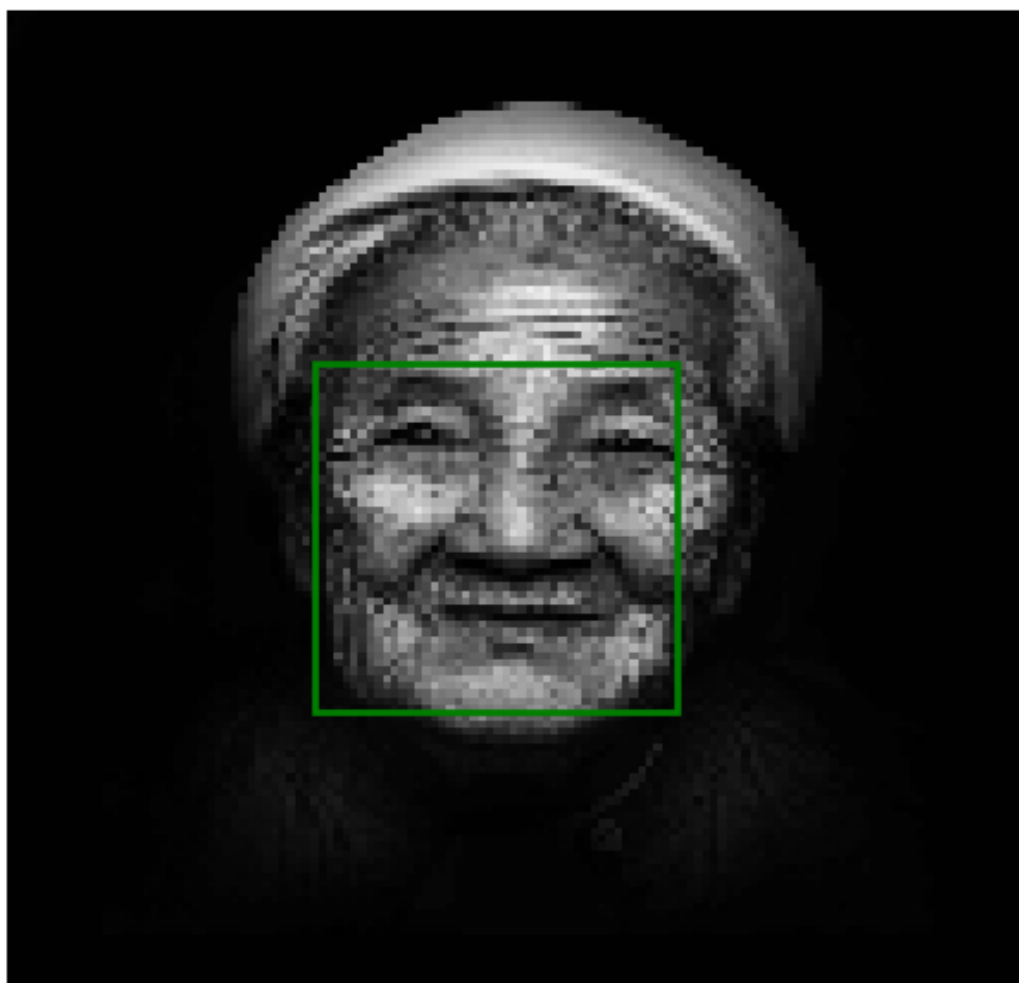
Plaintext: Model Prediction - Predicted Image

Green: Predicted | Red: Ground Truth



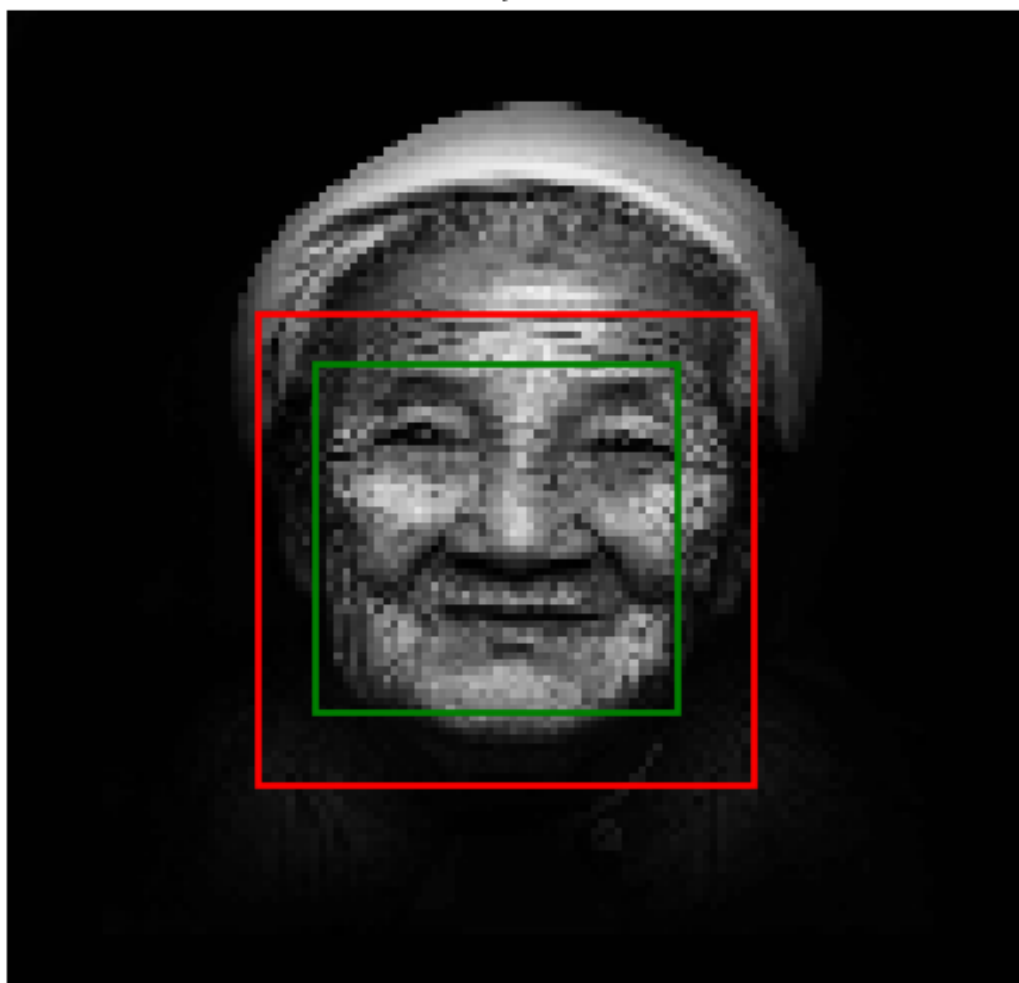
Plaintext: Model Prediction - Predicted Image

Green: Predicted



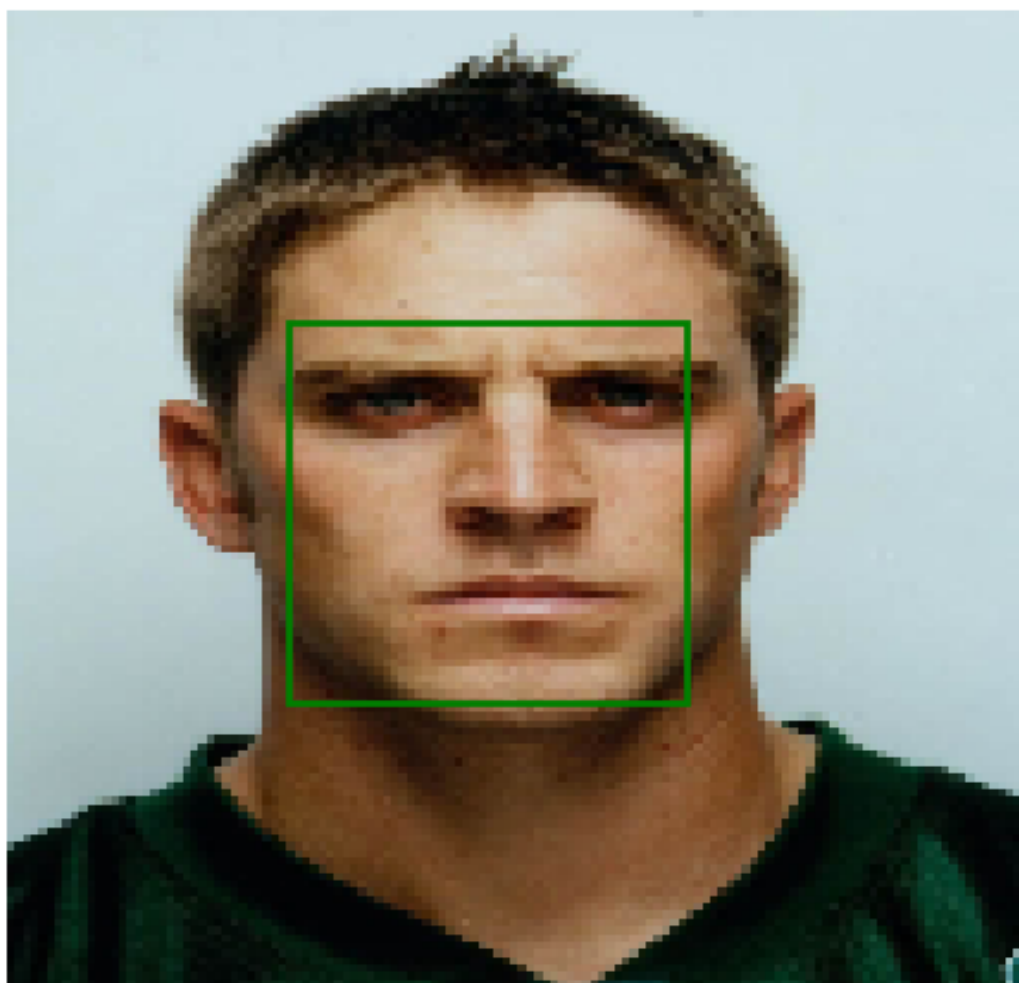
Plaintext: Model Prediction - Predicted Image

Green: Predicted | Red: Ground Truth



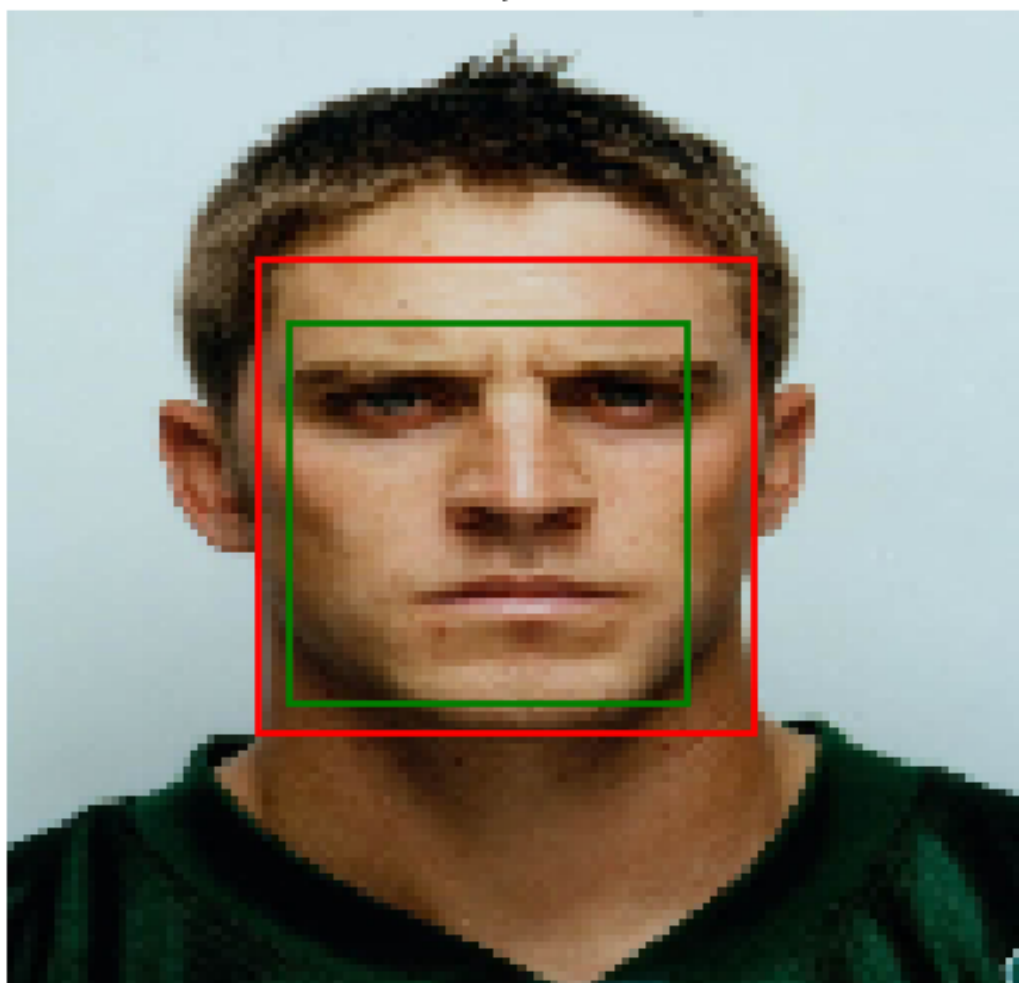
Plaintext: Model Prediction - Predicted Image

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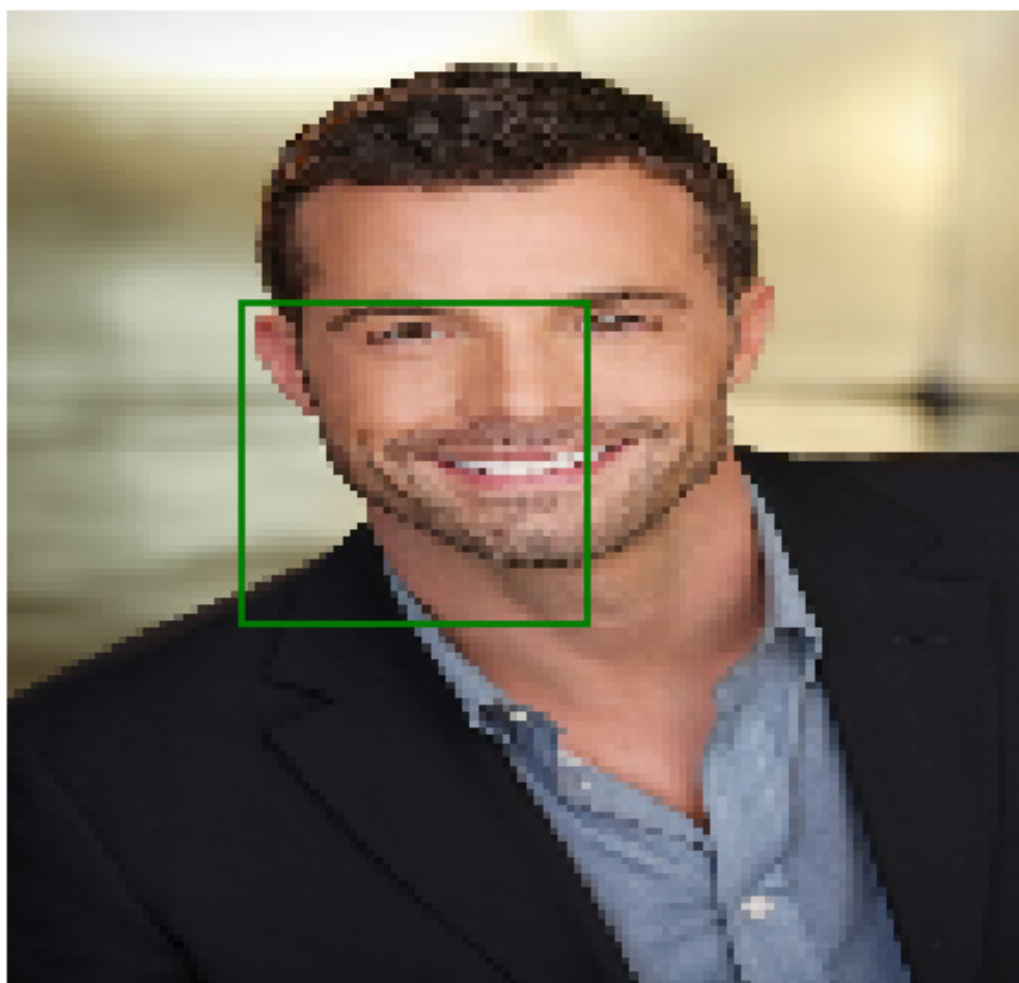
Plaintext: Model Prediction - Predicted Image

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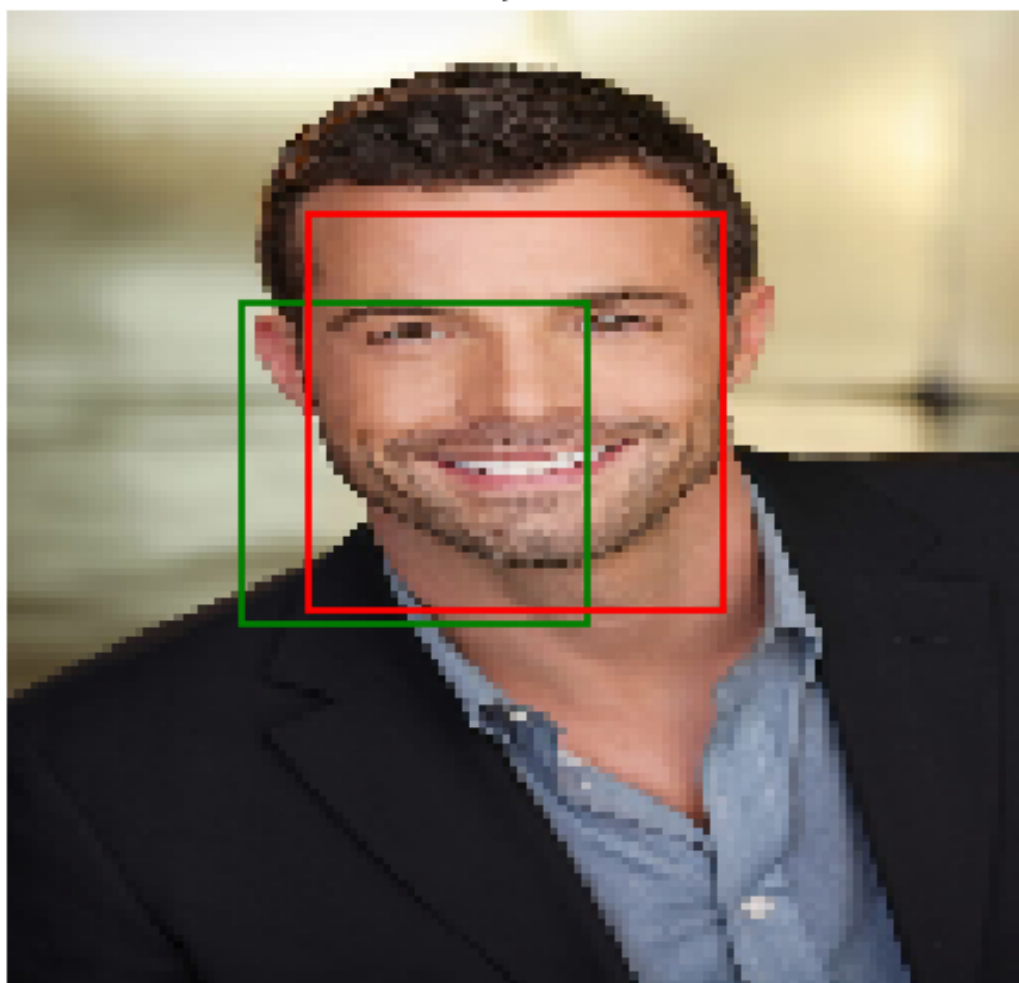
Plaintext: Model Prediction - Predicted Image

Green: Predicted



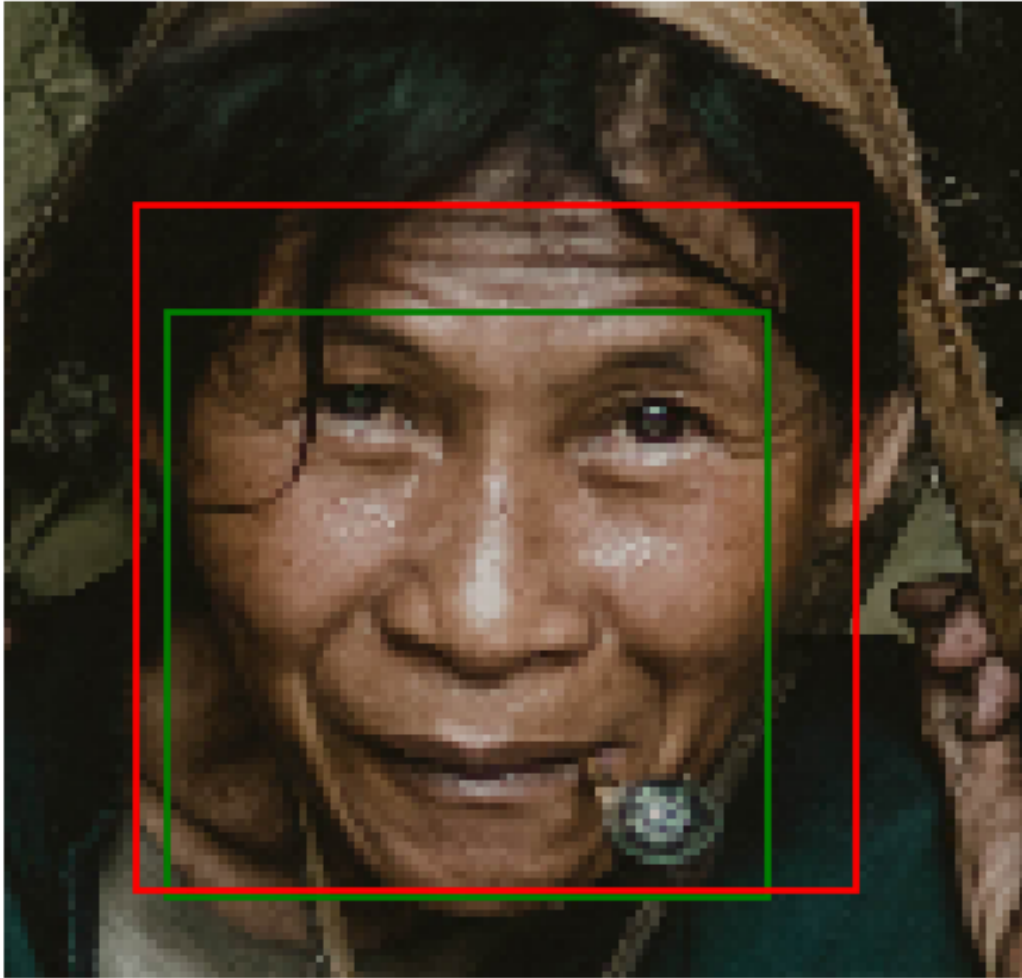
Plaintext: Model Prediction - Predicted Image

Green: Predicted | Red: Ground Truth



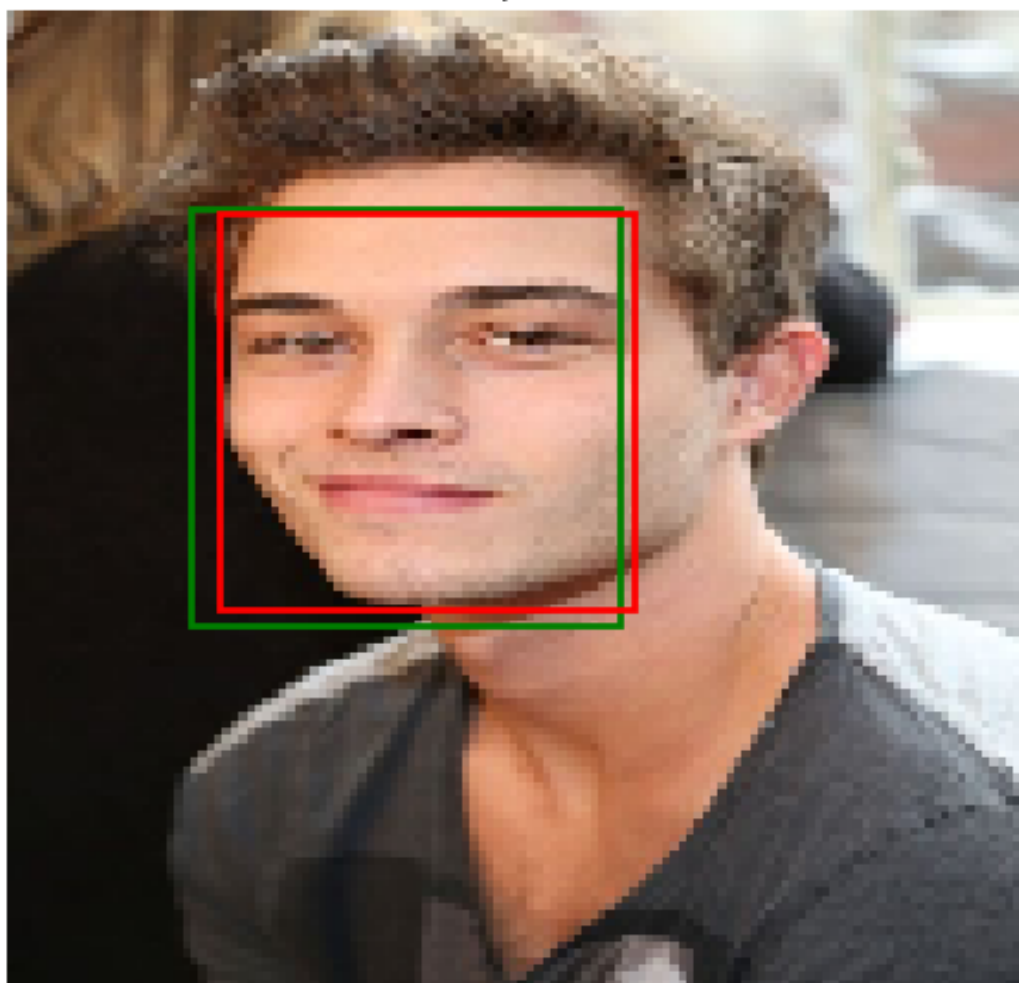
Encrypted: Model Prediction - Predicted Image

Green: Predicted | Red: Ground Truth



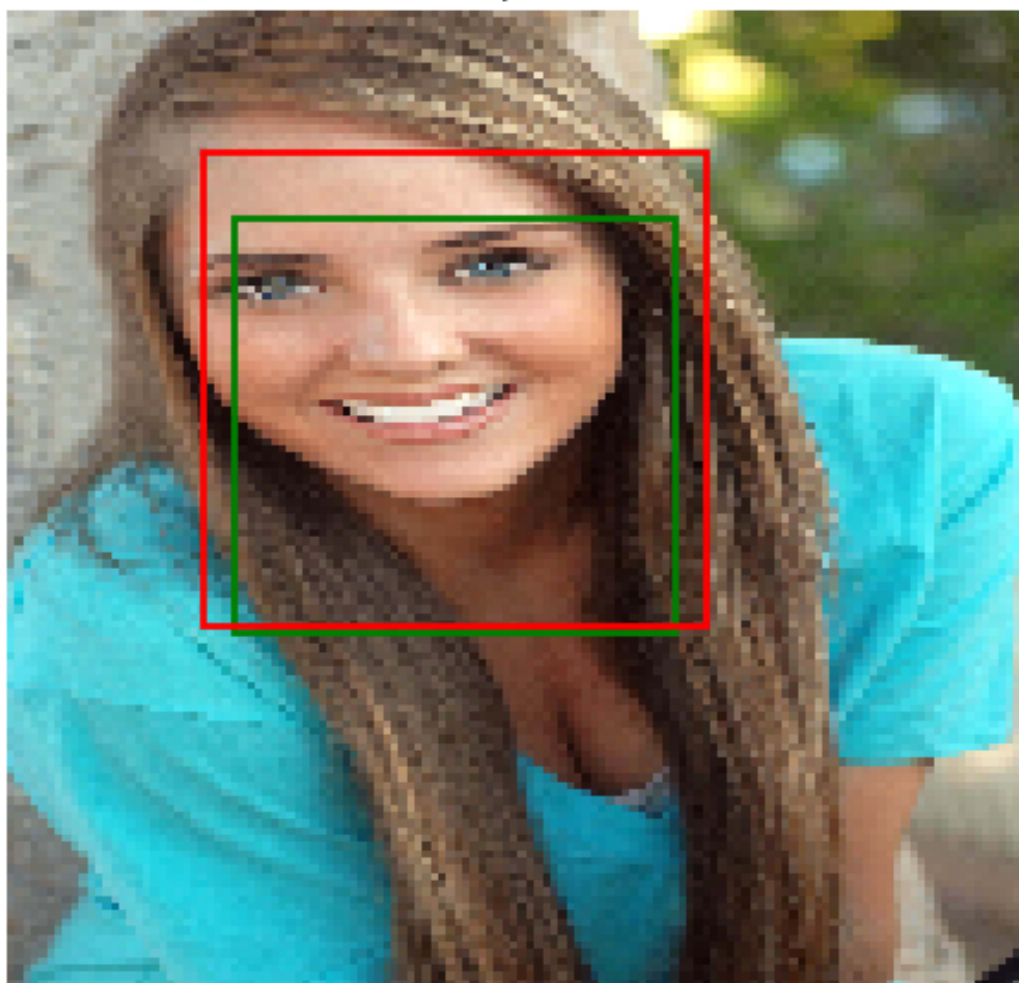
Encrypted: Model Prediction - Predicted Image

Green: Predicted | Red: Ground Truth



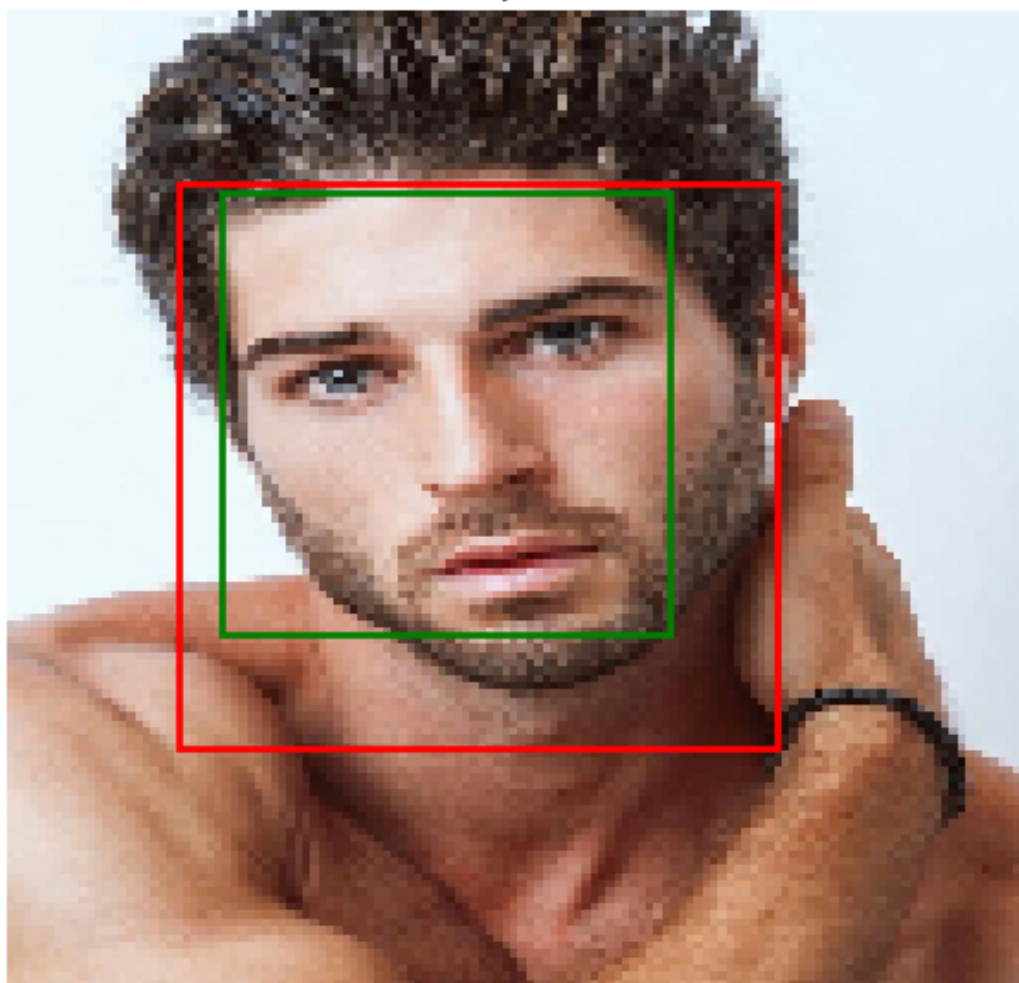
Encrypted: Model Prediction - Predicted Image

Green: Predicted | Red: Ground Truth



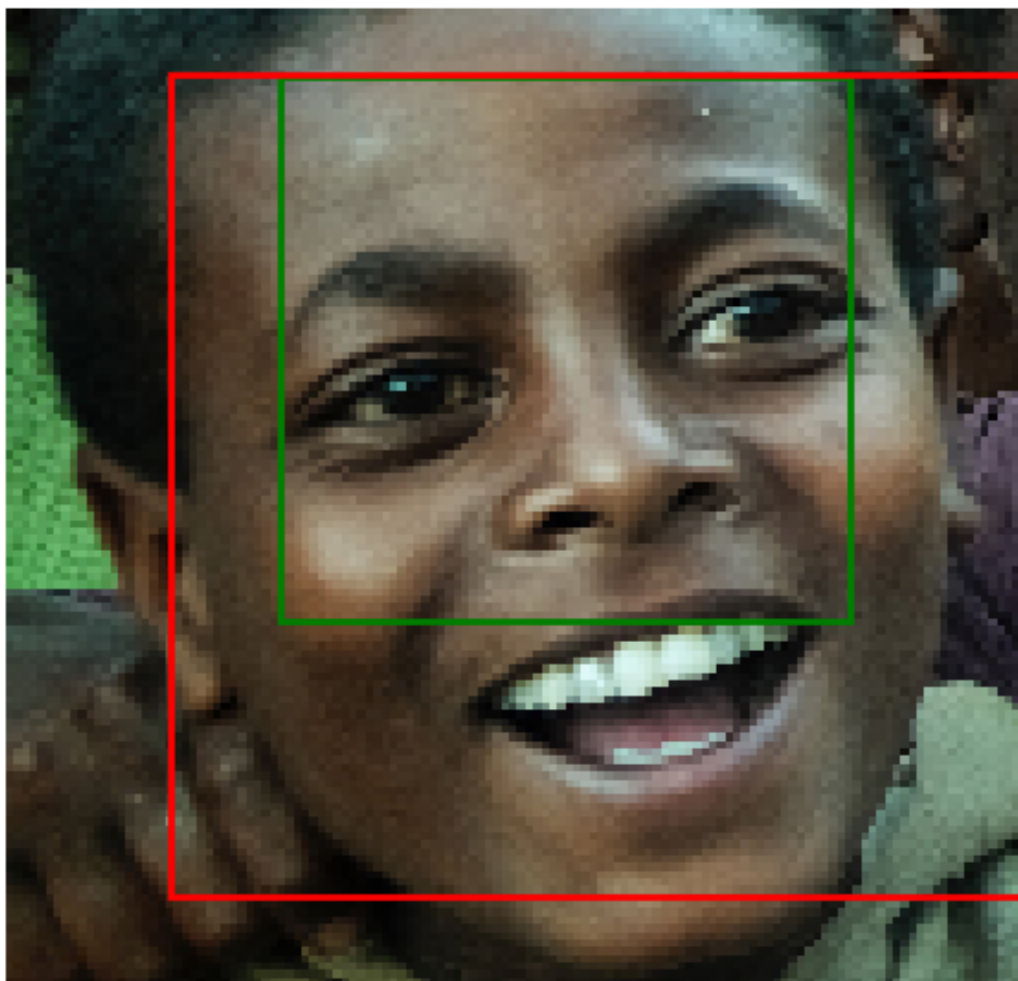
Encrypted: Model Prediction - Predicted Image

Green: Predicted | Red: Ground Truth



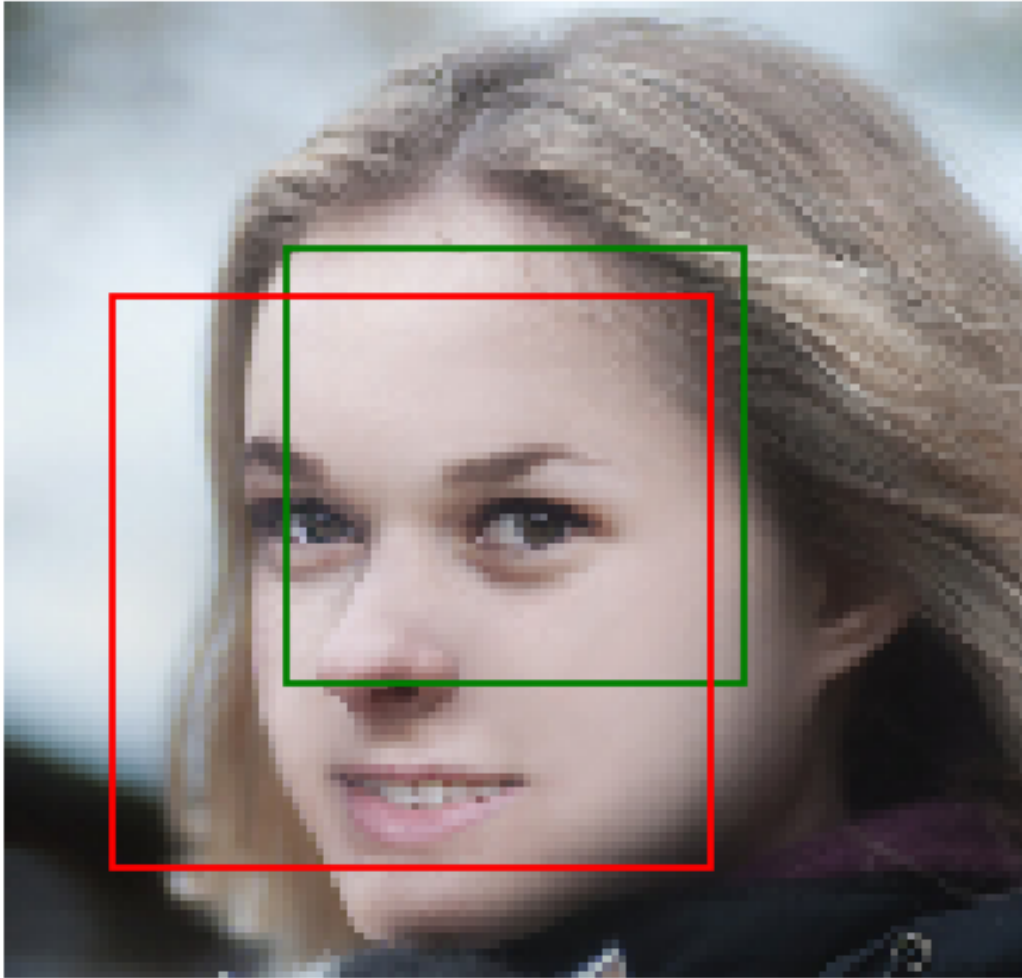
Encrypted: Model Prediction - Predicted Image

Green: Predicted | Red: Ground Truth



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Green: Predicted | Red: Ground Truth



Runtime System Resource Usage

encrypted-fhe: OS: macOS-26.0-arm64-arm-64bit
encrypted-fhe: Python Version: 3.10.8
encrypted-fhe: Processor: arm
encrypted-fhe: CPU Usage (%): 11.2
encrypted-fhe: RAM Usage (Used GB): 6.16
encrypted-fhe: RAM Usage (Total GB): 18.0
encrypted-fhe: RAM Usage (%): 77.5
encrypted-fhe: GPU: Not Available